

Article

Citizen-Science Data Reveal Global Diversity Patterns in the Gecko Genus *Hemidactylus*

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Abstract

Understanding how species richness is distributed across space is a central goal of macroecology and biogeography. The gecko genus *Hemidactylus* ranks among the most speciose and broadly distributed of all squamate genera, with species spanning tropical and subtropical environments on most continents. Yet the large-scale structure of its diversity has rarely been examined at a global scale. Here we assembled georeferenced occurrence data for the genus from the Global Biodiversity Information Facility (GBIF Access data is 14 March 2026) and, following cleaning and quality control, retained 143,858 records covering 157 currently recognized species (approximately 79% of the genus) to characterize broad-scale biodiversity patterns. Our analyses addressed the latitudinal diversity gradient, the distribution of species range sizes, continental richness, spatial clustering of diversity, and regional beta diversity. Richness declined steadily from the tropics toward the poles, producing a pronounced latitudinal diversity gradient (Pearson $r = -0.71$, $p < 0.001$). A generalized additive model captured a strong non-linear association between latitude and richness (deviance explained = 98.1%), consistent with tropical regions acting as principal centers of diversity. Range-size distributions were markedly right-skewed, with the majority of species restricted to comparatively small areas; however, these estimates co-varied strongly with per-species sampling effort ($r = 0.90$) and warrant cautious interpretation. We found only a weak, marginally significant association between range size and latitudinal midpoint, offering little support for Rapoport's rule in the genus. At the continental scale, *Hemidactylus* diversity was overwhelmingly concentrated in Afro-Asian tropical regions (Asia, 93 species; Africa, 66), whereas Europe and Oceania supported few species (four each). Spatial analyses indicated significant autocorrelation in richness (Moran's $I = 0.56$, $p < 0.001$) and well-defined diversity hotspots in East Africa, the Indian subcontinent, and Southeast Asia, and regional beta diversity was high (Jaccard dissimilarity up to 0.97), reflecting pronounced faunal turnover among continents.



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1. Introduction

Understanding how biological diversity is distributed in space is a central goal of ecology and biogeography. Identifying the processes that generate broad-scale diversity

patterns is important both for understanding what shapes species distributions and for anticipating how communities will respond to environmental change. One of the most widely documented of these patterns is the latitudinal diversity gradient (LDG), the tendency for species richness to increase from the poles toward the equator. Recorded across a broad range of taxa and ecosystems, the LDG is among the most general macroecological patterns known [1–4]. Its underlying causes nonetheless remain debated, with proposed explanations including latitudinal differences in speciation and extinction, the comparatively high climatic stability of the tropics, and the greater ecological opportunity afforded by warm environments [2,5,6].

Beyond latitudinal richness itself, macroecologists routinely interrogate geographic range size and compositional turnover to characterize how diversity is partitioned across space. A long-standing prediction in this area is Rapoport's rule, which holds that high-latitude species should occupy larger geographic ranges than their tropical counterparts [7]. Empirical tests have yielded inconsistent results across taxa, implying that range-size dynamics emerge from an interplay of climatic variability, dispersal capacity, and evolutionary history [1].

Reptiles, and geckos in particular, provide a useful system for investigating global macroecological patterns. Geckos are one of the most diverse lizard lineages and occur across a wide range of habitats and climatic zones; their richness is concentrated in tropical and subtropical regions, reflecting long evolutionary histories in warm climates [8]. Within this group, the genus *Hemidactylus* is one of the most species-rich and widely distributed of all reptile genera. Members of *Hemidactylus* inhabit tropical and subtropical landmasses as well as numerous oceanic islands, from Southeast Asia and Africa to the Americas [9]. Several species are highly successful colonizers, having achieved pantropical distributions through a combination of natural dispersal and human-mediated introduction [10–12].

Despite this broad distribution and ecological prominence, the genus has seldom been subjected to global-scale analysis of its diversity. Most prior work has concentrated on regional faunas, invasion dynamics, or phylogenetic relationships rather than worldwide macroecological structure. Consequently, several basic questions remain open: whether the genus conforms to the classical latitudinal diversity gradient, how its range sizes are arranged geographically, and how species composition shifts among major biogeographic realms. The macroecological literature on Old-World squamates has indeed been dominated by regionally focused niche-modeling and phylogeographic studies of comparatively narrow taxonomic or spatial scope [6,13–22], such that global syntheses for hyperdiverse, cosmopolitan reptile genera like *Hemidactylus* remain conspicuously scarce.

The growing accessibility of large biodiversity repositories, foremost among them the Global Biodiversity Information Facility (GBIF), now makes it feasible to interrogate broad-scale diversity directly from occurrence data. Such resources have become a mainstay of macroecological research into worldwide patterns of richness, range size, and compositional turnover across diverse taxa.

Building on a large compilation of georeferenced occurrence records, the present study undertakes a global biogeographic appraisal of the gecko genus *Hemidactylus*, organized around four questions:

- (1) Does *Hemidactylus* diversity follow a latitudinal diversity gradient?
- (2) Do species geographic ranges vary with latitude, as predicted by Rapoport's rule?
- (3) How is species richness distributed across continents and geographic regions?
- (4) To what extent do *Hemidactylus* assemblages differ among continents, indicating regional beta diversity and faunal turnover?

By jointly examining the spatial structure of richness, range size, and compositional turnover, the study delivers an integrated global characterization of diversity in one of the most widely ranging reptile genera.

2. Materials and Methods

2.1. Occurrence Data

Global occurrence records for the genus *Hemidactylus* were obtained from the Global Biodiversity Information Facility (GBIF; <https://www.gbif.org>). Data were accessed using the *rgbif* R package (3.5.8 version) [23], which provides an interface to the GBIF API. Only records containing geographic coordinates were retained. Records flagged by GBIF as having geospatial issues were excluded from the dataset. The final dataset used in the analyses, after the cleaning and quality-control steps described in Section 2.2, consisted of 143,858 occurrence records representing 157 *Hemidactylus* species. Although the genus currently comprises ~200 recognized species [8], the remaining taxa were excluded because they either lacked any georeferenced occurrence in GBIF or were represented by fewer than three valid records after filtering, an insufficient sample for inclusion in grid-based macroecological analyses. The species sampled here therefore cover approximately 79% of currently recognized diversity within the genus and span all major continental and insular regions in which *Hemidactylus* is known to occur. The complete list of the 157 species included in the analyses, together with the per-species number of occurrence records, number of occupied grid cells, estimated range size, IUCN Red List threat status (where available), and the set of countries in which each species occurs, is provided in Supplementary Table S1. Of the 143,858 occurrence records analyzed, 74.8% were classified by GBIF as HUMAN_OBSERVATION and approximately 89,889 records (62.5% of the dataset) were sourced from the iNaturalist Research-Grade Observations dataset as published through GBIF—that is, records for which GBIF’s publisher/institution-code metadata identifies iNaturalist as the contributing citizen-science platform, not a natural history museum or formal research institution—with the remainder originating from museum specimens and other GBIF-aggregated databases. This composition reflects the increasing contribution of research-grade citizen-science platforms to occurrence data for widespread reptiles, and the implications of this composition for inferring large-scale diversity patterns are discussed in Section 4.6. Roughly half of all currently recognized *Hemidactylus* species have been described in the past 25 years [8]; older GBIF records linked to the genus may therefore in some cases correspond to species that have subsequently been split or re-described, a potential source of nomenclatural uncertainty that should be borne in mind when interpreting per-species record counts. Species with fewer than three valid records after filtering were excluded from the grid-based analyses, but their identities and the corresponding record counts are nevertheless retained in Supplementary Table S1 so that readers can directly see which taxa were excluded and why.

All analyses were conducted in the R statistical environment [24].

2.2. Data Cleaning

Occurrence data were cleaned using the *CoordinateCleaner* R package (3.0.1 version) [25]. This procedure removes records that are likely to represent spatial errors, including records located at country centroids, biodiversity institutions, equal latitude and longitude coordinates, and zero coordinates. After cleaning, records lacking species identification or representing duplicate occurrences were removed from the dataset. Additionally, records with coordinate uncertainty greater than 10 km were excluded to reduce potential spatial errors in occurrence locations. These steps target geolocation and duplication errors and are not designed to detect biologically anomalous but geographically valid coordinates;

consequently, a small number of isolated occurrences from clearly extralimital localities (e.g., Canada, England, Belgium, Germany, Russia, and New Zealand), almost certainly reflecting escaped or released captive individuals, remain in the cleaned dataset and are visible as outlier points in Figure 1.

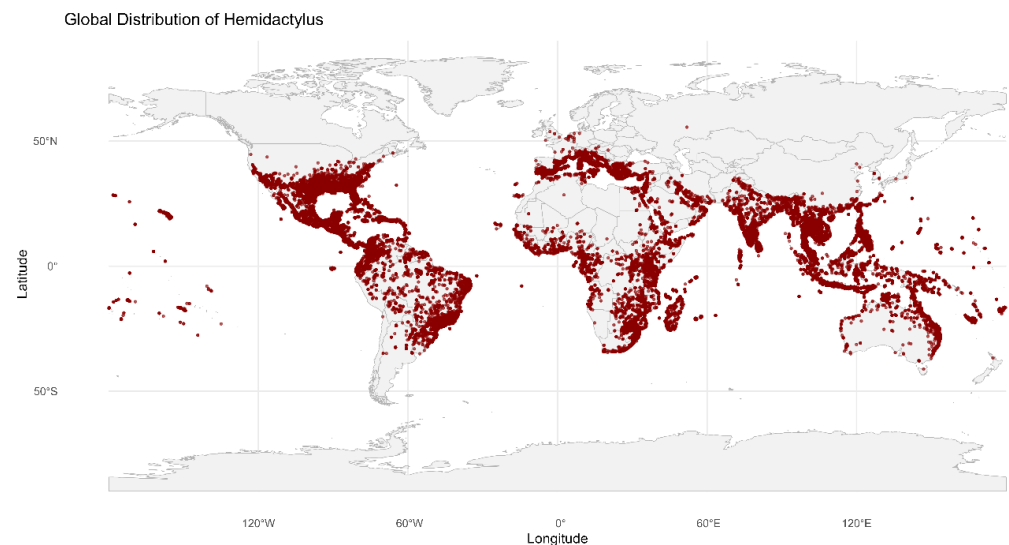


Figure 1. Global occurrence map of *Hemidactylus* species. Geographic distribution of *Hemidactylus* occurrence records obtained from the Global Biodiversity Information Facility (GBIF). Each point represents a georeferenced occurrence record after data cleaning and quality filtering. The map illustrates the global distribution of the genus across tropical and subtropical regions. A small number of isolated occurrence points fall well outside the genus's native and established introduced range (e.g., single records from Canada, England, Belgium, Germany, Russia, and New Zealand); these almost certainly represent escaped or released captive individuals rather than established populations and were not removed by automated cleaning, since they do not meet any coordinate-error or duplicate-record flagging criterion (Section 2.2).

2.3. Spatial Data Processing

Spatial operations were conducted using the *sf* R package (0.6-3 version) [26], which implements simple features for spatial vector data in R. Occurrence records were converted to spatial objects using geographic coordinates (WGS84).

Continental boundaries were obtained from the *rnatuarearth* dataset using the *rnatuarearth* R package (0.1.0 version) [27]. Occurrence points were spatially joined to continental polygons in order to assign each record to a continent.

2.4. Latitudinal Diversity Gradient

Species richness along latitudinal gradients was calculated by grouping occurrence records into 5° latitudinal bands. Species richness was defined as the number of unique species within each band. Data manipulation was conducted using the *dplyr* R package (1.1.4 version) [28].

The relationship between latitude and species richness was evaluated using Pearson correlation analysis.

Because latitudinal diversity gradients often exhibit non-linear patterns, a generalized additive model (GAM) was fitted using the *mgcv* R package (1.8-22 version) with thin-plate regression splines. The smooth term was specified with the default basis dimension ($k = 10$) and an automatically selected smoothing parameter, which allowed flexible estimation of non-linear relationships between latitude and species richness while penalizing excessive wiggleness [29]. The choice of 5° latitudinal bands as the analytical unit balances biolog-

ical resolution against per-band sample size and follows established practice in global macroecological assessments of vertebrate diversity gradients [4,30].

2.5. Species Range Size Estimation

Species range sizes were estimated using a grid occupancy approach, a commonly used method in macroecology [31]. A global raster grid with 1° spatial resolution was generated using the terra R package (1.7-65 version) [32]. Occurrence records were assigned to raster cells, and the number of occupied cells per species was calculated.

Species range size was estimated as the number of occupied 1° grid cells multiplied by the nominal cell area at the equator (~12,321 km²). This approach assumes equal cell areas across latitudes, which slightly overestimates the contribution of high-latitude cells because the true geodesic area of a 1° × 1° cell decreases with increasing latitude (e.g., ~6200 km² at 60°). The implications of this approximation are addressed in the Discussion, and equal-area projection alternatives are noted as a methodological refinement for future work.

2.6. Testing Rapoport's Rule

To test Rapoport's rule, the relationship between species range size and the latitudinal midpoint of species distributions was evaluated using Pearson correlation analysis.

The latitudinal midpoint of each species distribution was calculated as the mean latitude of all occurrence records [7]. To explicitly partition any latitudinal range-size signal from a sampling-intensity confound, the Pearson correlation was complemented with a multiple linear regression in which species-level log-transformed range size was modelled as a function of (i) absolute latitudinal midpoint and (ii) log-transformed per-species occurrence records (a proxy for sampling effort). The partial regression coefficient and significance of the latitudinal-midpoint term, conditional on sampling effort, were used as the effort-adjusted test of Rapoport's rule.

2.7. Continental Species Richness

Species richness per continent was calculated using occurrence records assigned to continental regions. Richness was defined as the number of unique species recorded within each continent.

2.8. Global Species Richness Hotspots

Global patterns of species richness were mapped using a grid-based richness approach. Species richness per grid cell was calculated and visualized using the ggplot2 R package (2.2.1 version) [33]. Richness maps were used to identify geographic hotspots of *Hemidactylus* diversity.

2.9. Spatial Autocorrelation

Spatial clustering of species richness was evaluated using Moran's I statistic, implemented in the spdep R package (0.7-9 version) [34]. Moran's I tests whether spatial patterns differ significantly from random distributions. Spatial weights were generated using an eight-nearest-neighbour ($k = 8$) approach, which corresponds to a queen-style contiguity for regular grids and is widely used as a default in global gridded macroecological studies [30,34].

2.10. Regional Beta Diversity

Differences in species composition among continents were quantified using Jaccard dissimilarity [35]. Pairwise dissimilarity matrices were computed with the vegan R package (2.6-6 version) [36] and visualized as a heatmap with hierarchical clustering implemented in the pheatmap R package (1.0.12 version) [37]. To disentangle the contributions of species

replacement and species loss to overall compositional differentiation, total Jaccard dissimilarity was decomposed into its turnover and nestedness-resultant components following the Baselga [38,39] framework, implemented in the betapart R package (1.6.1 version) [40]. Within this framework, the turnover component (β_{jt}) represents species replacement independent of richness differences, while the nestedness-resultant component (β_{jne}) captures the proportion of dissimilarity explained by richness differences when one assemblage is a subset of another.

2.11. Sampling Bias Analysis

To evaluate potential sampling bias, the relationship between sampling effort (number of occurrence records) and species richness was analyzed using linear regression models. To reduce potential sampling bias caused by uneven occurrence records, spatial thinning was applied using a grid-based filtering approach. Occurrence records were aggregated into 0.1° spatial cells, and only one record per species per grid cell was retained. This approach reduces spatial clustering of occurrence records and has been widely used in large-scale macroecological analyses. Spatial thinning was applied primarily to evaluate the robustness of large-scale diversity patterns. Latitudinal diversity analyses were conducted using both the original and thinned datasets, while continental richness and beta diversity analyses were performed using the original dataset to retain complete species occurrence information. In addition to spatial thinning, sampling effort was incorporated as a covariate in confirmatory regression models. For the latitudinal diversity gradient, band-level species richness was modelled as a function of absolute latitude and log-transformed per-band sampling effort, so that the latitude coefficient in the adjusted model represents variation in richness across latitude after accounting for differences in occurrence density. The same covariate-adjusted regression framework was applied at the species level for Rapoport's rule, with log-transformed species range size modelled as a function of absolute latitudinal midpoint and log-transformed per-species occurrence records (see Section 2.6). This regression-based correction was selected as the most parsimonious and interpretable approach for the present descriptive macroecological context, in which the goal is to characterize large-scale patterns rather than to model species-level detection or occupancy processes.

Statistical models were implemented using base R functions.

2.12. Software and Packages

All analyses were conducted in R version 4.4.1 [24]. Key R packages used in this study included rgbif [23], CoordinateCleaner [25], rnatrualearth [27], dplyr [28], mgcv [29], terra [32], ggplot2 [33], spdep [34], vegan [36], and pheatmap [37].

3. Results

3.1. Global Occurrence Patterns

A total of 143,858 occurrence records representing 157 *Hemidactylus* species were retained from GBIF following data cleaning and quality control. Mapped occurrences confirmed the genus's extensive spread across tropical and subtropical zones, with the densest concentrations in Africa, South Asia, and Southeast Asia (Figure 1).

Records were highly unevenly distributed across species. Four broadly introduced taxa (*Hemidactylus frenatus*, *H. mabouia*, *H. turcicus*, and *H. brookii*) accounted for a large proportion of records. These species alone represented approximately 81% of all occurrence records and, summed across species, approximately 59% of the total species-level range area (and an identical 59% of all occupied grid cells), reflecting the strong influence of cosmopolitan synanthropic species in global occurrence databases. They therefore dominate

the dataset in both sampling volume and geographic coverage, although this range-area figure should be read in light of the strong confound between estimated range size and per-species sampling effort documented below (Section 3.4). Because these four taxa contribute the overwhelming majority of records and, more importantly, account for virtually all of the introduced, human-mediated occurrences in the dataset, whereas the remaining narrow-range species are predominantly native endemics with negligible introduced records the native-versus-introduced distinction was assessed in detail for these four species, where it is most consequential for interpreting global patterns. The global distribution of each of these four taxa is shown separately in Supplementary Figure S1, in which occurrence records are classified as falling within or outside the native extant range of each species according to IUCN range polygons. These maps make clear that a substantial fraction of the records for all four taxa particularly in the Americas, where the genus has no native populations lie outside the inferred native range and thus represent introduced or human-mediated occurrences.

To mitigate spatial sampling bias arising from clustered records, we applied a grid-based thinning procedure, aggregating occurrences into 0.1° cells and keeping a single record per species per cell. This step trimmed the dataset from 143,858 to 26,989 records without changing the species count (157 species), confirming that the reduction chiefly eliminated spatially redundant observations rather than removing species.

3.2. Latitudinal Diversity Gradient

Species richness exhibited a clear latitudinal diversity gradient, rising in the tropics and tapering toward higher latitudes. A single latitudinal band attained a maximum of 38 species, whereas high-latitude bands held only a handful. Pearson correlation confirmed a strong negative latitude-richness association ($r = -0.71$, $p < 0.001$), richness decreasing with increasing absolute latitude. A generalized additive model (GAM) further revealed a strong non-linear relationship between latitude and species richness ($\text{edf} = 8.27$, $F = 68.27$, $p < 0.001$), explaining 98.1% of deviance in species richness (adjusted $R^2 = 0.97$); the high effective degrees of freedom, approaching the upper bound permitted by $k = 10$, indicate substantial deviation from a simple linear trend. Richness clearly peaked in tropical latitudes and declined toward both hemispheric extremes (Figure 2). After spatial thinning to reduce sampling bias, the latitudinal diversity gradient remained strong and statistically significant ($r = -0.78$, $p < 0.001$), indicating that the gradient persists once spatial clustering of records is reduced. To evaluate this pattern more conservatively, a multiple linear regression was fitted with band-level species richness as the response variable and both absolute latitude and log-transformed sampling effort (occurrence records per latitudinal band) as predictors. Latitude remained a significant predictor of species richness ($\beta = -1.02$, $p = 0.002$), whereas sampling effort was not significant once latitude was accounted for ($p = 0.915$; Figure 3). These results indicate that the latitudinal diversity gradient detected in *Hemidactylus* is not an artefact of uneven sampling intensity.

3.3. Invasive Species Sensitivity Analysis

To evaluate the potential influence of widely introduced species on global diversity patterns, a sensitivity analysis was conducted excluding four cosmopolitan taxa (*H. frenatus*, *H. mabouia*, *H. turcicus*, and *H. brookii*). Even after removing these species, the latitudinal diversity gradient remained strong ($r = -0.73$, $p < 0.001$), indicating that the global richness pattern is not driven solely by human-mediated dispersal.

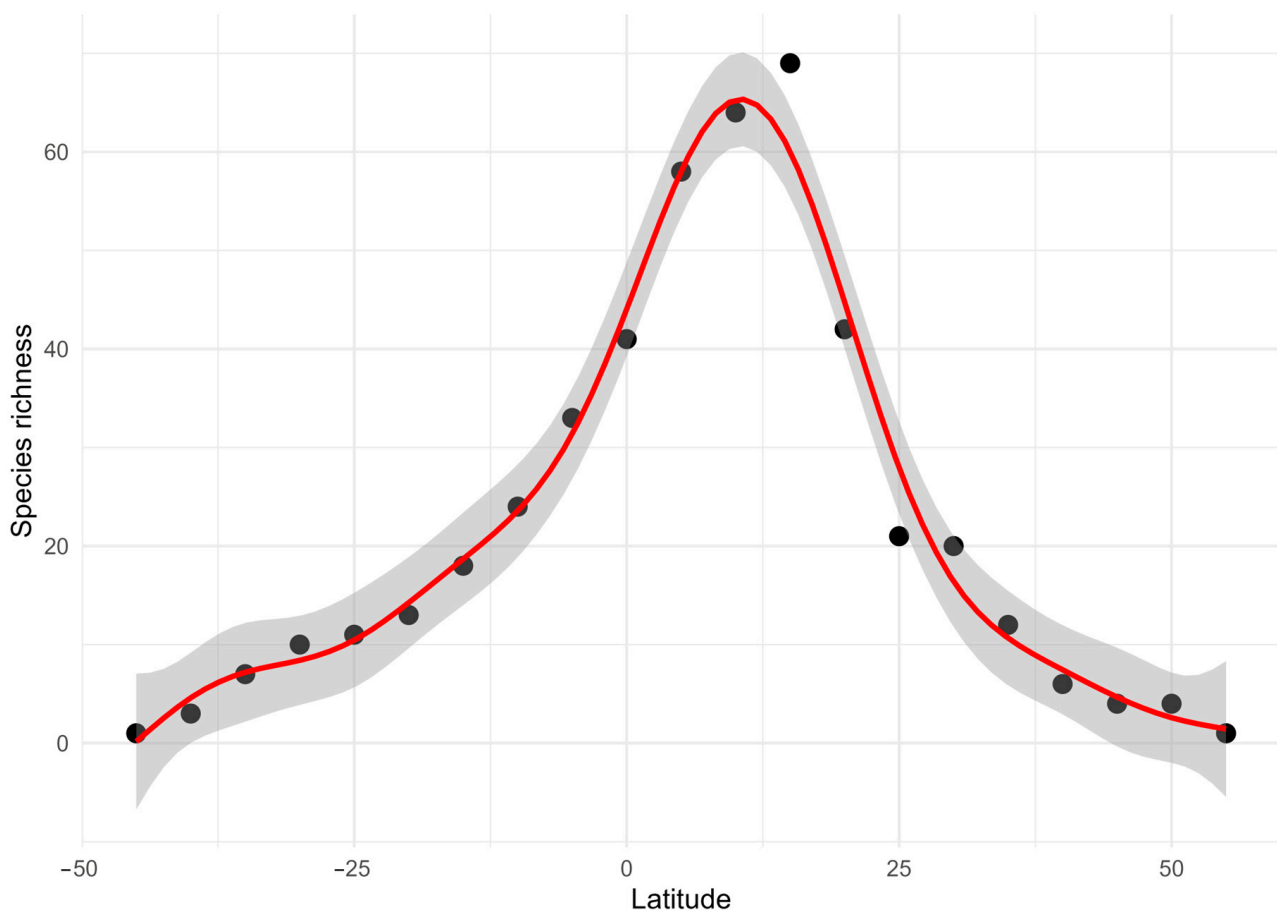


Figure 2. Latitudinal diversity gradient of *Hemidactylus* species richness. Species richness across latitudinal bands (5° intervals). Points represent observed richness values, while the red curve represents the fitted generalized additive model (GAM). Richness peaks in tropical latitudes and declines toward higher latitudes.

3.4. Species Range Size Patterns

Estimated species range sizes varied widely among *Hemidactylus* species. The median estimated range size was approximately 36,963 km², while a small number of species exhibited extremely large ranges exceeding 17 million km². The range-size distribution was strongly right-skewed, indicating that most species occupy relatively small geographic ranges while only a few species are widely distributed. However, a very strong positive relationship was detected between estimated range size and the number of occurrence records per species ($r = 0.90$, $p < 0.001$; Figure 4), implying that the observed range-size estimates are heavily confounded with per-species sample size. Inferences regarding absolute range sizes should therefore be interpreted with caution; the observed right-skew is qualitatively consistent with a genuine biological signal documented across taxa [41], but the magnitude of estimated ranges particularly for poorly sampled species likely reflects a mixture of true geographic extent and sampling intensity.

3.5. Testing Rapoport's Rule

Species range size was compared with the latitudinal midpoint of species distributions to evaluate Rapoport's rule. The analysis revealed a weak positive relationship between range size and absolute latitudinal midpoint ($r = 0.15$), but the relationship was marginal and did not reach conventional statistical significance ($p = 0.064$). Given the strong co-variation between range estimates and sampling effort documented above ($r = 0.90$), the absence of a clear latitudinal trend should not be interpreted as conclusive evidence against

Rapoport's rule in *Hemidactylus*; rather, it indicates that any latitudinal range–size signal in this genus is at most weak and is not detectable once a substantial sampling-effort confound is taken into account. To address this confound explicitly, a multiple linear regression was fitted with log-transformed species range size as the response variable, absolute latitudinal midpoint as the explanatory variable, and log-transformed per-species occurrence records as a covariate representing sampling effort. Sampling effort was a highly significant predictor of estimated range size ($p < 0.001$), whereas latitudinal midpoint showed no significant independent effect after controlling for occurrence density ($p = 0.828$). These results indicate that the apparent variation in estimated range size is strongly governed by sampling intensity rather than by latitudinal position, providing no robust support for Rapoport's rule in *Hemidactylus* once sampling effort is taken into account.

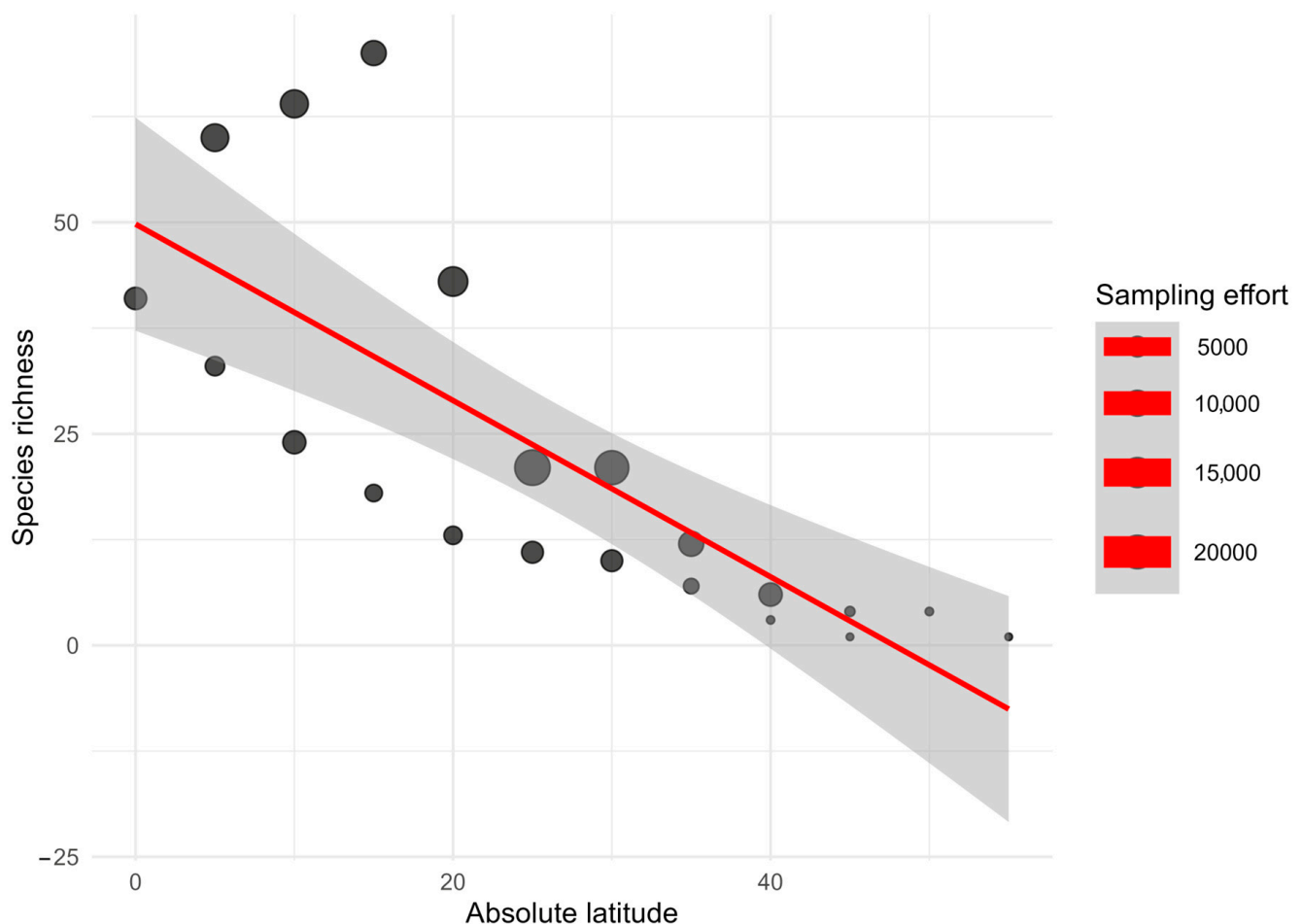


Figure 3. Latitudinal diversity gradient of *Hemidactylus* species richness with sampling effort included as a covariate. Points represent observed species richness within 5° latitudinal bands, with point size proportional to per-band sampling effort (number of occurrence records, see legend). The red line represents the fitted multiple linear regression ($\text{richness} \sim |\text{latitude}| + \log(\text{sampling effort})$) and the shaded grey envelope shows the 95% confidence interval around the latitude effect. After accounting for sampling effort, absolute latitude remained a significant predictor of species richness ($\beta = -1.02$, $p = 0.002$), whereas sampling effort was not significant ($p = 0.915$), indicating that the latitudinal diversity gradient is not an artefact of uneven sampling intensity.

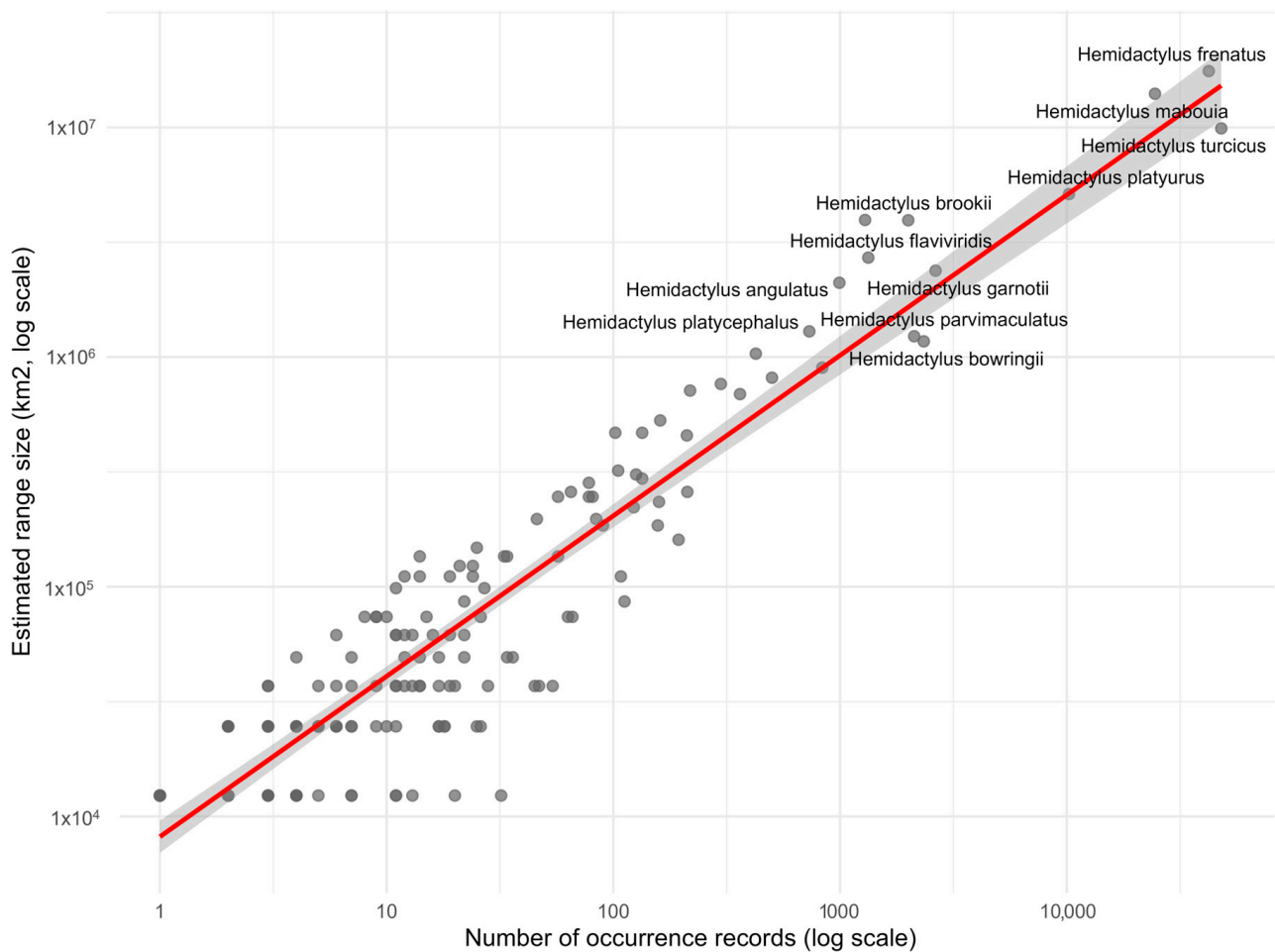


Figure 4. Relationship between estimated range size and number of occurrence records across *Hemidactylus* species. Both axes are shown on logarithmic scales. Species with the largest estimated ranges are labeled. The strong positive relationship indicates that estimated range size is strongly influenced by sampling intensity.

3.6. Continental Patterns of Species Richness

Continental richness was strongly unequal. Asia led with 93 species, ahead of Africa with 66; by contrast, the Americas held comparatively few (North America = 12 species, South America = 10 species), and Europe and Oceania each supported just four (Figure 5). This distribution identifies the tropical Afro-Asian region as the diversity core of the genus *Hemidactylus*.

3.7. Global Species Richness Hotspots

Spatial mapping of species richness revealed clear geographic hotspots of *Hemidactylus* diversity. The highest richness values were concentrated in East Africa, the Indian subcontinent, and Southeast Asia, while temperate regions exhibited markedly lower richness (Figure 6).

3.8. Spatial Autocorrelation

Moran's I analysis revealed strong spatial autocorrelation in species richness (Moran's $I = 0.56$, $p < 0.001$), indicating that *Hemidactylus* diversity is spatially clustered rather than randomly distributed. This confirms the presence of geographically structured diversity hotspots.

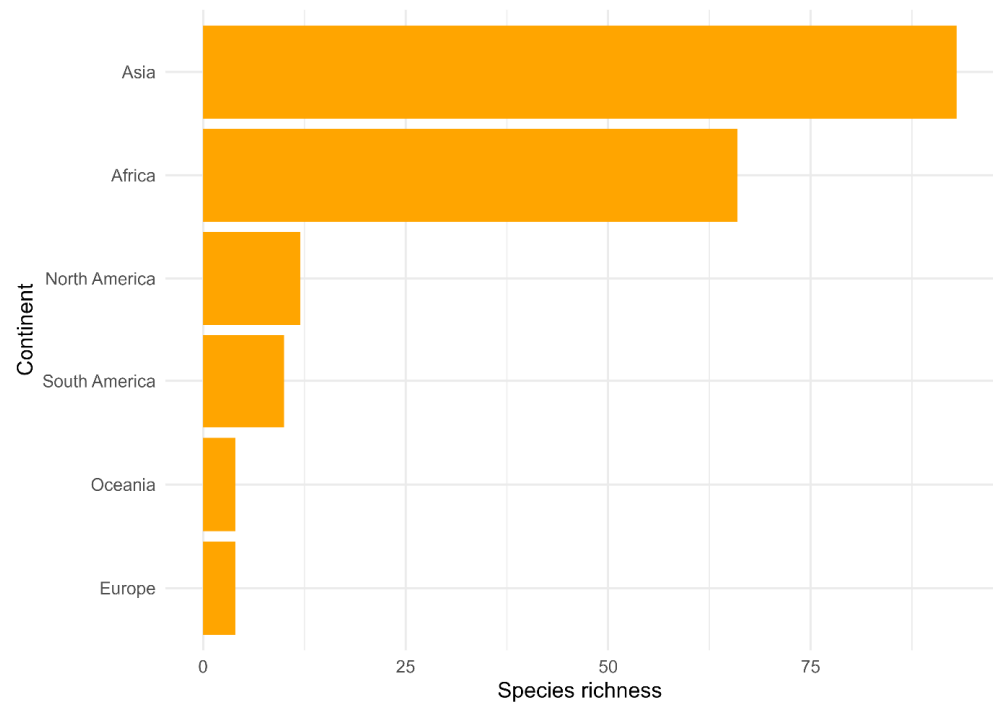


Figure 5. Continental species richness of *Hemidactylus*. Number of *Hemidactylus* species recorded on each continent based on GBIF occurrence data. Asia and Africa harbor the highest species richness, while Europe and Oceania contain relatively few species. Country-level richness, which is highly uneven within continents (e.g., more than 50 species in India versus only a few species in Cambodia, Thailand, or Indonesia), is shown in Supplementary Figure S2. Continental totals are as follows: Asia, 93 species; Africa, 66; North America, 12; South America, 10; Europe, 4; Oceania, 4.

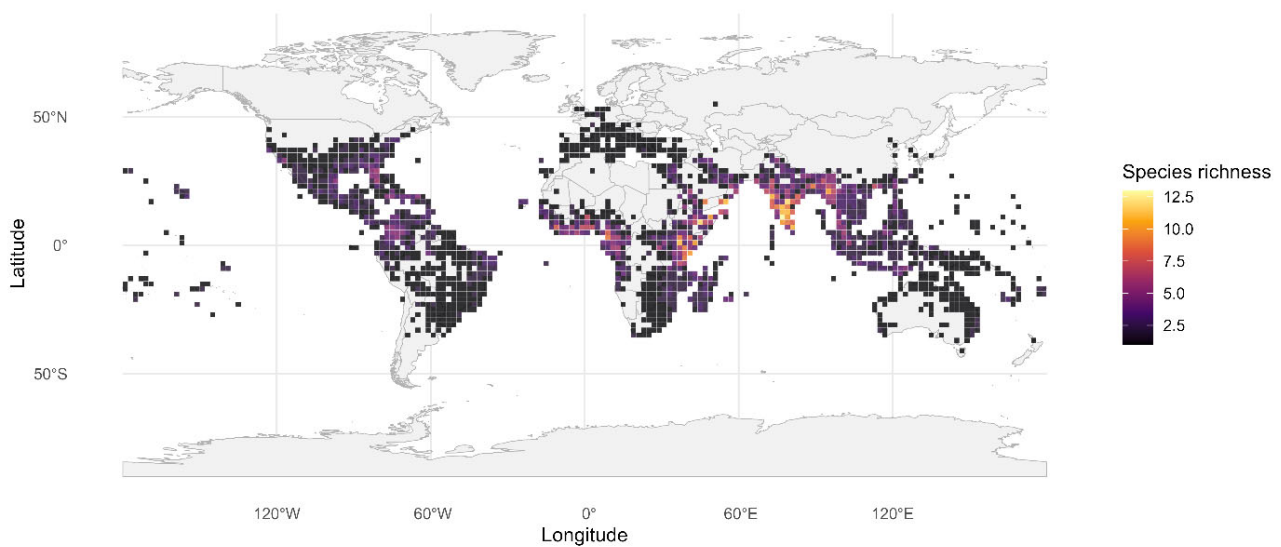


Figure 6. Global species richness hotspots. Global spatial distribution of *Hemidactylus* species richness based on 1° grid-cell richness analysis. Warmer colors indicate higher species richness. Major diversity hotspots are concentrated in East Africa, the Indian subcontinent, and Southeast Asia.

3.9. Regional Beta Diversity

Regional beta diversity analysis revealed high levels of faunal turnover among continents. Jaccard dissimilarity values were generally high, ranging from 0.91 to 0.97 between Afro-Asian and other regions, indicating strong compositional differences in *Hemidactylus* assemblages worldwide. In contrast, North and South America exhibited relatively lower

dissimilarity (0.53), suggesting more similar species compositions between these regions (Figure 7).

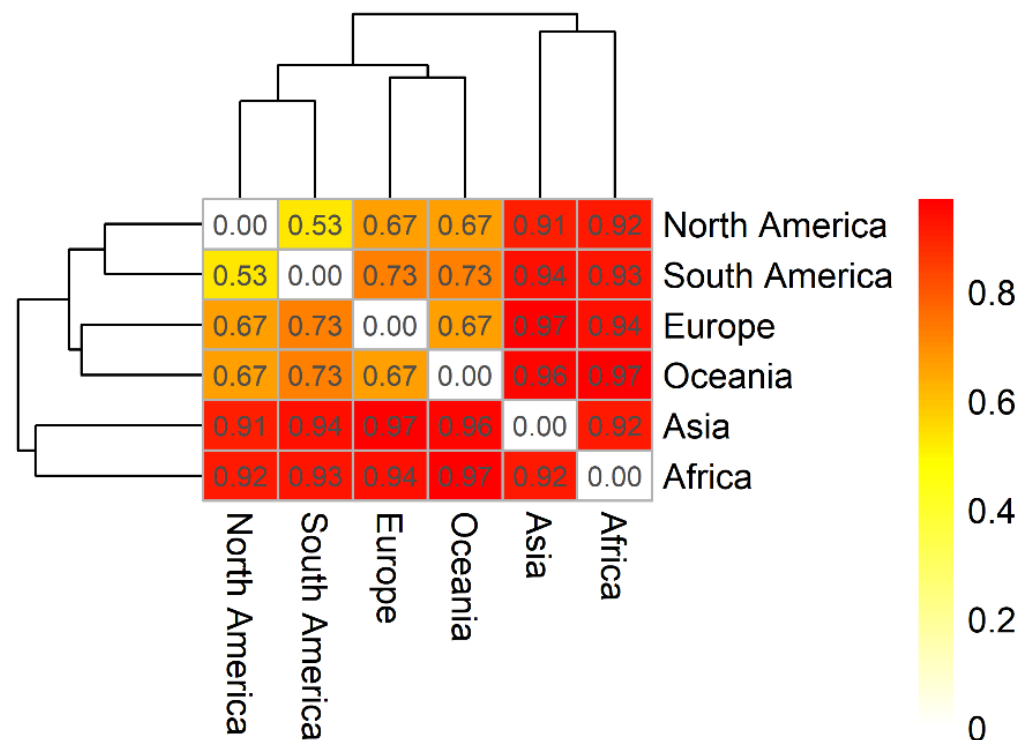


Figure 7. Regional beta diversity and hierarchical clustering of *Hemidactylus* assemblages among continents. Heatmap showing pairwise Jaccard dissimilarity values representing differences in species composition among continents. Warmer colors indicate higher faunal dissimilarity. Hierarchical clustering based on the Jaccard distance matrix reveals two major assemblage groups, with the American regions (North and South America) clustering together, while Afro-Eurasian and Oceanian regions form a separate cluster. The results indicate high global faunal turnover in *Hemidactylus* assemblages, with particularly strong compositional differences between the Americas and the Afro-Asian regions.

Decomposition of beta diversity revealed that continental differences were driven by both species' turnover and nestedness components. Turnover values were particularly high between Africa and Asia (0.90), indicating strong faunal replacement between these regions. In contrast, high nestedness values were observed between Europe and Africa (0.94), suggesting that European assemblages represent subsets of more species-rich tropical faunas.

3.10. Relationship Between Sampling Effort and Species Richness

Sampling effort varied across latitudes and was positively associated with species richness. Linear regression analysis revealed a weak but significant relationship between sampling effort and species richness ($R^2 = 0.20$, $p = 0.043$). However, sampling effort explained only a small proportion of the observed variation in species richness, suggesting that the global diversity pattern largely reflects biological rather than sampling processes (Figure 8).

Importantly, the major diversity patterns remained consistent after spatial thinning of occurrence records.

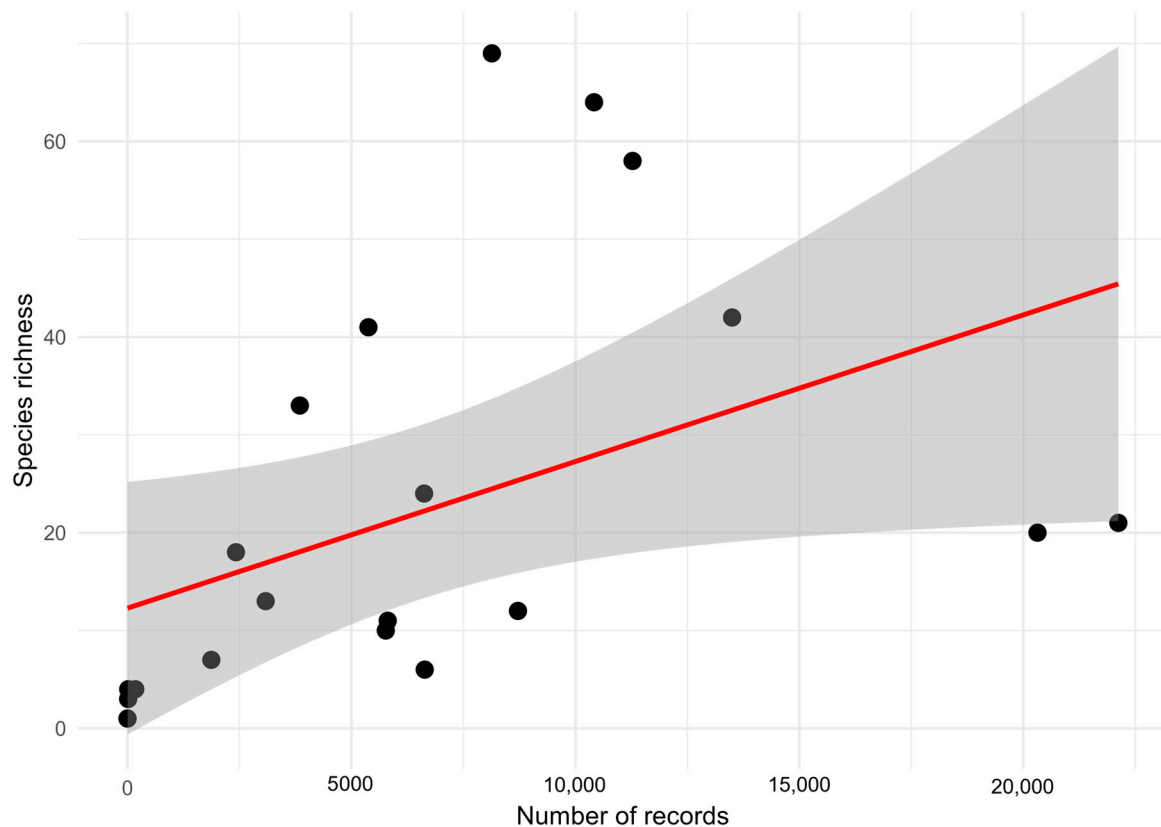


Figure 8. Band-level relationship between the number of occurrence records and species richness. Each point represents a single 5° latitudinal band; the x-axis is the total number of GBIF occurrence records summed across all species occurring within that band (i.e., per-band sample size), and the y-axis is the number of species recorded in the same band. The red line represents the linear regression model fitted across bands, indicating a weak but significant positive relationship between per-band sample size and per-band species richness.

4. Discussion

4.1. Global Diversity Patterns in *Hemidactylus*

This work offers an integrated, global-scale biogeographic appraisal of the gecko genus *Hemidactylus*. Across the analyses, three large-scale patterns were consistently recovered: a marked latitudinal diversity gradient, pronounced spatial clustering of richness, and substantial faunal turnover between continents.

The clearest of these is the latitudinal diversity gradient (LDG), with richness reaching its maximum in the tropics and declining toward higher latitudes. As one of the most general patterns in ecology, the LDG has been documented in plants, vertebrates, and invertebrates [1,2], and our findings are consistent with earlier global reptile analyses identifying the tropics as the region of peak reptile diversity [30]. The strongly non-linear latitude-richness relationship recovered by the GAM further supports the view that tropical environments act as diversification centers for many ectothermic groups [6].

Several non-exclusive mechanisms could underpin the pronounced tropical richness observed in *Hemidactylus*. Relative to temperate zones, tropical regions tend to combine greater climatic stability with higher energy availability, a pairing argued to promote elevated speciation and depressed extinction [2,42,43]. The ectothermic physiology of reptiles, moreover, ties their performance closely to ambient temperature, a dependence that may further amplify the tropical concentration of diversity [6,21,30].

4.2. Geographic Range Size Patterns and Rapoport's Rule

Estimated geographic range sizes varied substantially across *Hemidactylus* species, the resulting distribution being strongly right-skewed: most species were confined to comparatively small ranges, while only a handful spanned very large areas. Right-skewed range-size distributions of this kind are a familiar macroecological signature documented across many taxa [41].

Contrary to the predictions of Rapoport's rule [7], the present study found no significant relationship between species range size and latitude. This lack of support for Rapoport's rule is consistent with many previous studies that have reported weak or inconsistent support for the hypothesis, particularly in ectothermic organisms [1]. Several explanations have been proposed for this inconsistency, including differences in dispersal ability, climatic tolerance, and historical biogeographic processes.

Importantly, the present study detected a very strong positive relationship between estimated range size and sample size (i.e., the number of occurrence records per species) ($r = 0.90$), which raises a fundamental interpretive concern: occurrence-based range estimates likely reflect a substantial admixture of true geographic extent and per-species observation count rather than biological extent alone. Such sampling effects have been widely documented in studies using global biodiversity databases and highlight the importance of accounting for sampling bias when interpreting biogeographic patterns [31]. Two consequences follow. First, the absolute range-size estimates reported here (particularly the median, which is heavily influenced by data-poor species) should not be over-interpreted, and ratio comparisons across species or biogeographic regions warrant caution. Second, the absence of a significant Rapoport effect must be read against the backdrop of this confound: lineages with few records, if any, would have systematically underestimated ranges, which could mask a Rapoport-type signal even if one were biologically present. To address this confound explicitly, a multiple linear regression including the log-transformed number of records per species as a covariate was fitted alongside the Pearson correlation (Section 3.5). After accounting for occurrence density, the number of records per species emerged as a highly significant predictor of estimated range size ($p < 0.001$), whereas latitudinal midpoint showed no significant independent effect ($p = 0.828$). This adjusted analysis confirms that the apparent variation in estimated range size is governed primarily by per-species sample size rather than by latitudinal position, and that no robust evidence for Rapoport's rule is recovered once the number of records is controlled for. More sophisticated occupancy-based estimators (e.g., Bayesian hierarchical models partitioning observation and process variance) could be applied in future work, but for the descriptive biogeographic scope of the present study the regression-based correction with the number of records as a covariate provides a parsimonious and interpretable solution.

The latitudinal midpoint of species distributions was used to test Rapoport's rule, a widely applied approach in macroecological studies because it reduces biases associated with uneven sampling across species ranges. Midpoint methods provide a more stable representation of species distribution centers compared with approaches based solely on minimum or maximum latitude values.

Because grid cell area varies with latitude, range size estimates based on fixed-degree grids may slightly underestimate ranges at higher latitudes. However, this effect is unlikely to substantially influence the broad macroecological patterns detected here.

4.3. Continental Diversity Patterns

The results show that *Hemidactylus* diversity is strongly concentrated in Afro-Asian tropical regions, with Asia and Africa harboring the majority of species. This pattern is consistent with the evolutionary history of the genus, which likely originated in the Old

World tropics and subsequently diversified across Africa and Asia [9]. The relatively low species richness observed in Europe and Oceania likely reflects climatic constraints and historical dispersal limitations, mirroring the narrow climatic envelopes and projected range contractions reported for cool-temperate and montane reptiles at the periphery of the Western Palearctic [20].

In contrast, the Americas contain relatively few *Hemidactylus* species, despite the large geographic area of these continents. This pattern may reflect a combination of historical biogeographic barriers and relatively recent colonization events. Several species present in the Americas are known to have expanded their ranges through human-mediated dispersal, particularly synanthropic species such as *Hemidactylus frenatus* and *H. mabouia* [10]. Comparable evidence of putative human-mediated dispersal extending the realized range of squamates beyond their historical limits has recently been reported for other Western Palearctic lizards [11], reinforcing the idea that anthropogenic transport is a recurrent driver of contemporary distributions in synanthropic reptiles. Additional examples illustrate how diverse traits favor human-mediated dispersal in geckos. The Mediterranean gecko *Hemidactylus turcicus*, native to the circum-Mediterranean region, is now widely established across the southern United States and parts of Mexico, where it persists in commensal urban habitats far from its native range. The parthenogenetic *H. garnotii* is particularly noteworthy: its all-female reproductive mode allows successful establishment from a single propagule, dramatically lowering the demographic barrier to anthropogenic colonization, and it is now widespread across tropical regions globally. The same biological feature underlies the extreme invasive success of the unrelated parthenogenetic gecko *Lepidodactylus lugubris*, which has spread across the Pacific, the Americas, and parts of Africa, illustrating that the combination of synanthropic habitat use and reduced reproductive barriers represents a recurrent ecological syndrome favoring long-distance human-mediated dispersal in geckos.

The sensitivity analyses excluding widely introduced species showed that the latitudinal diversity gradient remained strong even after removing the four most cosmopolitan taxa. This indicates that the observed diversity gradient is not solely driven by human-mediated dispersal. Instead, the persistence of the pattern suggests that natural biogeographic processes continue to structure global diversity patterns in *Hemidactylus*.

Phylogenetic studies suggest that the genus originated in the Old-World tropics and subsequently expanded through multiple dispersal events across Africa and Asia [9], consistent with the diversity patterns observed in the present study.

4.4. Regional Beta Diversity and Faunal Turnover

The analysis of regional beta diversity revealed high levels of compositional turnover among continents, indicating that *Hemidactylus* assemblages differ markedly among major geographic regions. Such high beta diversity is consistent with patterns observed in many global reptile groups and likely reflects historical isolation and regional diversification processes [6,16,30].

Interestingly, the Americas exhibited lower faunal dissimilarity between North and South America compared with other continental comparisons. This pattern suggests relatively similar assemblages within the New World, likely driven by shared colonization history and the presence of a limited number of widely distributed species.

High beta diversity among continents highlights the importance of regional evolutionary histories in shaping present-day biodiversity patterns. Geographic barriers, historical dispersal events, and regional diversification likely played major roles in structuring the global distribution of *Hemidactylus* rather than recent local adaptation alone documented across diverse vertebrate groups, including both squamates and mammals [15,44].

The decomposition of beta diversity further revealed that continental differences were driven by both species' turnover and nestedness components. High turnover between Africa and Asia suggests substantial faunal replacement between these regions, likely reflecting independent diversification histories. In contrast, high nestedness values between Europe and Africa indicate that temperate assemblages represent subsets of more diverse tropical faunas.

4.5. Spatial Clustering and Biodiversity Hotspots

Spatial analyses revealed strong spatial autocorrelation in species richness, indicating that *Hemidactylus* diversity is not randomly distributed across the globe but instead forms clear geographic clusters. Global richness hotspots were concentrated in East Africa, the Indian subcontinent, and Southeast Asia, regions that are widely recognized as major biodiversity centers for reptiles and other vertebrate groups [30].

These regions combine several factors favorable to high reptile diversity, including warm climates, heterogeneous habitats, and long-term climatic stability, with topographic and habitat heterogeneity repeatedly identified as key correlates of reptile occupancy at regional scales [19]. Such conditions may promote both speciation and persistence of lineages over evolutionary timescales [2,6].

4.6. Limitations and Implications

Although occurrence-based biodiversity datasets provide powerful opportunities for biogeographic research, they also introduce potential biases related to uneven sampling effort. In this study, the number of records per species explained part of the variation in species richness and range size estimates, indicating that some observed patterns may be influenced by differences in sampling intensity among regions. A related consideration concerns the composition of the GBIF dataset itself. As reported in Section 2.1, 74.8% of the occurrence records analyzed here were classified as human observations and approximately 62.5% (~89,889 records) were contributed through the iNaturalist platform, as identified by GBIF's publisher metadata, rather than by natural history museums or other research institutions. The dominance of citizen-science observations highlights both the strengths and the limitations of large biodiversity databases: although such datasets substantially improve geographic coverage relative to traditional museum-only inventories, they may also introduce spatial sampling biases associated with urbanization, accessibility, and observer activity [31]. These biases are likely to be most pronounced for widespread synanthropic species and most attenuated for narrow-range tropical endemics that occur far from human population centers, and they reinforce the need to interpret absolute richness and range-size values rather than the overall geographic patterns themselves with appropriate caution.

Nevertheless, the major biodiversity patterns detected in this study particularly the strong latitudinal diversity gradient and the concentration of diversity in tropical Afro-Asian regions are consistent with well-established macroecological patterns reported in other reptile groups [30]. However, several specific limitations should be emphasized when interpreting the present results. First, the GBIF occurrence dataset does not distinguish native from introduced records, and four widespread synanthropic species (*H. frenatus*, *H. mabouia*, *H. turcicus*, and *H. brookii*) collectively contribute approximately 81% of all occurrence records, with extensive introduced populations, particularly in the Americas, where no native *Hemidactylus* populations exist. For the four most pervasively introduced taxa, IUCN native-range polygons are available and were used to classify occurrence records as native or introduced (Supplementary Figure S1), confirming that a large fraction of their records especially in the Americas fall outside the native range. Restricting this

native/introduced assessment to these four species is not an arbitrary limitation but a deliberate design choice: together they contribute approximately 81% of all occurrence records and approximately 59% of the summed species-level range area, and they account for essentially all of the human-mediated occurrences in the dataset, so classifying them captures the dominant source of potential bias from introduced populations. The remaining 153 species are overwhelmingly narrow-range native endemics for which introduced records are negligible, and a genus-wide, record-level native/introduced filter could in any case not be applied because standardized global IUCN distribution polygons are currently unavailable for the majority of *Hemidactylus* species: the genus presently includes approximately 200 described species [8] with ongoing taxonomic revisions and uneven geographic coverage, and no unified native-range dataset currently exists that would permit consistent species-level filtering across the entire dataset analyzed here. The present study was therefore designed as a global occurrence-based macroecological analysis rather than a strictly native-range biogeographic synthesis, and the sensitivity analysis excluding the four cosmopolitan synanthropic taxa (Section 3.3) confirmed that the principal latitudinal diversity gradient is recovered even when the most pervasively introduced species are removed. Second, occurrence-based range-size estimates remain heavily confounded with per-species sampling effort ($r = 0.90$); the effort-adjusted multiple regression presented in Section 3.5 controls for this confound at the inferential level, but absolute range areas and in particular cross-species comparisons of range size should not be over-interpreted. Third, the original claim that the LDG was “robust to uneven sampling effort” on the basis of spatial thinning alone has been moderated: the inference that the gradient is not an artefact of sampling effort now rests on the covariate-adjusted multiple regression reported in Section 3.2, in which latitude remained a significant predictor of richness ($\beta = -1.02$, $p = 0.002$) after accounting for log-transformed sampling effort. Within these caveats, the patterns described here likely reflect genuine large-scale biodiversity structure in *Hemidactylus*, but should be interpreted as occurrence-based macroecological signal rather than as strictly native biogeographic range, and would benefit from further validation as comprehensive species-level native-range data become available for the genus.

4.7. Phylogenetic Considerations

A further limitation worth highlighting is that the analyses presented here treat species as independent units, without accounting for shared evolutionary history. Within a single hyperdiverse genus, species are by definition non-independent, and shared ancestry can generate apparent macroecological patterns that are not adaptive responses to current environments but rather reflect deep phylogenetic structure [5,45]. The *Hemidactylus* radiation is known to comprise several geographically structured clades with origins traceable to different Old-World source regions [9], and the most recent comprehensive molecular phylogeny of the genus [46] recovers an Indian-plate origin around 58 Myr ago followed by multiple out-of-India dispersals, with parallel in situ radiations subsequently establishing India and continental Africa as the two principal cradles of contemporary diversity. The strong concentration of richness in Afro-Asian tropics observed here is therefore at least partially attributable to the in situ diversification of these clades rather than to current climatic filtering alone. Similarly, the high beta diversity between the New World and the Afro-Asian regions almost certainly reflects the small number of independent colonization events into the Americas rather than fundamentally different climatic optima. The biogeographic reconstruction of Agashe et al. [46], an Indian origin, parallel India/Africa cradle of in situ radiation, and limited New-World invasion is therefore concordant with the continental and beta-diversity signals reported here, supporting the conclusion that the global geography of *Hemidactylus* richness has been structured by deep evolutionary history

as well as by contemporary climate. Future work integrating a time-calibrated phylogeny of *Hemidactylus* with the occurrence-based macroecological framework developed here through measures such as phylogenetic diversity, evolutionary distinctiveness, and tests of niche conservatism along a dated tree would allow a more mechanistic interpretation of the observed patterns and would help disentangle ecological from historical drivers of contemporary diversity.

5. Conclusions

To our knowledge, this represents the first global macroecological synthesis of diversity in the genus *Hemidactylus*. Collectively, the results show a marked tropical richness peak, strong spatial clustering of species richness, and high inter-continental beta diversity (i.e., pronounced compositional turnover between continents, with few species shared between the Afro-Asian and New-World faunas; Section 3.9). Taken together, they underscore the standing of the tropical Afro-Asian region as the principal center of *Hemidactylus* diversity and the joint imprint of historical and ecological processes on global reptile biodiversity.

Beyond corroborating a classical latitudinal diversity gradient in one of the most speciose reptile genera, our analyses pinpoint three regions, East Africa, the Indian subcontinent, and Southeast Asia, as the foremost global hotspots of *Hemidactylus* diversity. These regions overlap closely with broader hotspots of reptile endemism [30] and warrant prioritization in conservation planning, particularly given that many narrow-range species those occupying the small-range tail of the strongly right-skewed distribution are likely to be most vulnerable to habitat loss and climate change.

In light of accelerating anthropogenic climate change and ongoing range modification of squamates across the Western Palearctic and adjacent regions [20–22], the strong tropical concentration of *Hemidactylus* richness identified here implies that future anthropogenic change in tropical regions rather than in temperate latitudes will determine the fate of most of the genus's diversity. Two anthropogenic drivers in particular warrant attention. Projected tropical climate change, including increases in mean temperature and changes in the seasonality of precipitation, is expected to disproportionately affect tropical ectotherms whose thermal safety margins are narrow [47]. Equally important, however, are direct human pressures on tropical landscapes: continued human population growth, agricultural intensification, and habitat conversion in regions of high *Hemidactylus* endemism most notably in the Indian subcontinent, where more than 50 narrowly distributed species occur are likely to disproportionately threaten range-restricted species whose climatic niches are confined to small, increasingly fragmented areas. The combination of climate change and accelerating land-use change in tropical Asia and Africa therefore constitutes the most immediate threat to the global diversity of the genus, and conservation priorities should be assessed against the species-level range size and threat-status information compiled in Supplementary Table S1. The recurrent role of human-mediated dispersal in expanding the distribution of synanthropic taxa [10,11] further suggests that the realized geography of *Hemidactylus* will continue to diverge from its native climatic envelope, with consequences for invasion biology and biotic homogenization that warrant continued monitoring.

Three priorities emerge for future research. First, integrating a time-calibrated phylogeny would allow ecological versus historical drivers of diversity to be partitioned and would extend the present occurrence-based framework into a phylogenetically informed one. Second, formal bioregionalization analyses (e.g., network-based clustering of grid-cell faunas) are needed to test whether the continental-scale beta diversity patterns identified here translate into discrete, robustly supported biogeographic units. Third, projecting species-level climatic niches under future scenarios particularly for narrow-range, data-poor species in East Africa and South Asia would translate the macroecological patterns

described here into actionable conservation priorities. Together, these directions would consolidate the present synthesis and move from describing the global diversity of one of the most successful reptile radiations to explaining and forecasting it.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/d18070429/s1>, Supplementary Figure S1. Global occurrence records of four widespread synanthropic *Hemidactylus* species in relation to their native extant distributions based on IUCN range polygons. Red points represent GBIF occurrence records located within the native extant range, whereas blue points indicate records occurring outside the native extant distribution, likely reflecting introduced or human-mediated populations. The figure highlights the extensive anthropogenic expansion of *H. frenatus*, *H. mabouia*, *H. turcicus*, and *H. brookii* beyond their presumed native distributions. Supplementary Figure S2. Country-level species richness of *Hemidactylus* based on GBIF occurrence records. Colors indicate the total number of recorded species per country. White areas represent countries without available occurrence records. Supplementary Table S1. Summary of the global distribution dataset for *Hemidactylus* species, including the number of GBIF occurrence records, occupied grid cells, estimated geographic range size (km²), countries of occurrence, and IUCN Red List conservation status.

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